

Vasavi College of Engineering (A)
(Affiliated to Osmania University)

Hyderabad-500031

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Experiment 1:

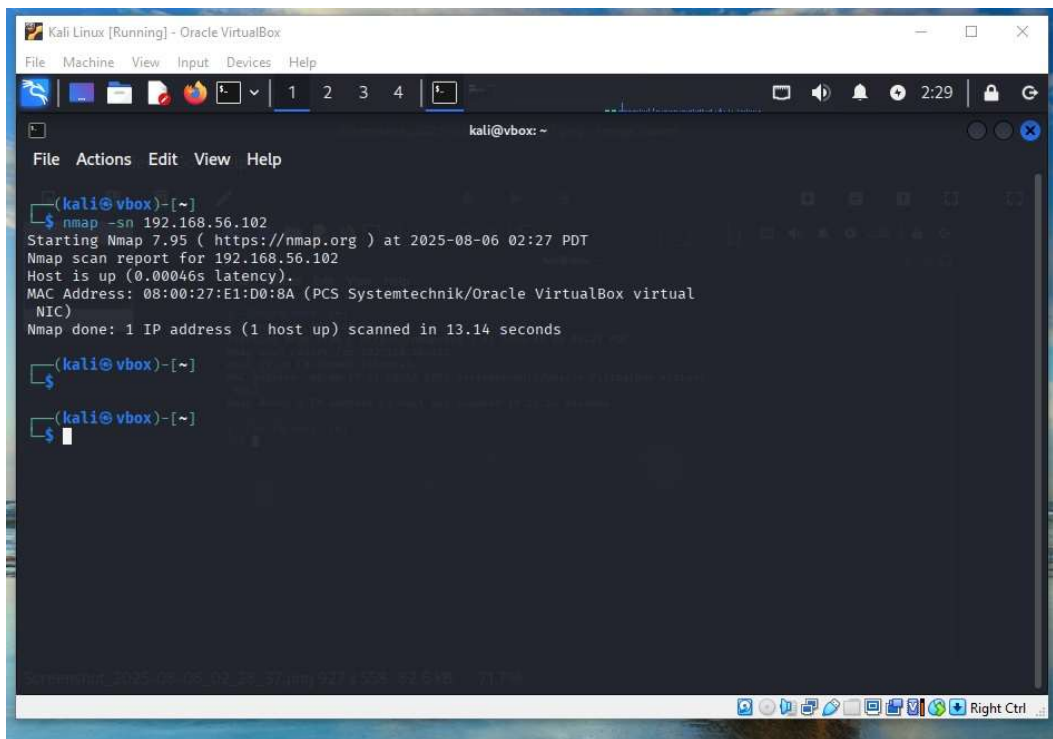
Network Scanning and Information Gathering: Perform host discovery, port scanning, and service enumeration.

1. Host Discovery (using Nmap):

Host discovery, or "ping scan," identifies active hosts on a network. Nmap can achieve this using various techniques: ARP Scan (for local networks).

Code

```
nmap -sn <target_subnet>
```

A screenshot of a Kali Linux terminal window running inside an Oracle VM VirtualBox. The terminal shows the execution of the command 'nmap -sn 192.168.56.102'. The output indicates that the host is up with a latency of 0.00046s and provides the MAC address 08:00:27:E1:D0:8A, identifying it as a PCS Systemtechnik/Oracle VirtualBox virtual NIC. The scan was completed in 13.14 seconds. The terminal window has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. The top of the window shows 'Kali Linux [Running] - Oracle VirtualBox' and a taskbar with various icons and a system clock showing 2:29. The bottom of the window shows a taskbar with icons for applications and a 'Right Ctrl' button.

Example:

This command performs an ARP scan to discover live hosts on the local network. Ping Scan (for remote networks).

Code: `nmap -PE -PS443 -PA80 <target_IP_range>`

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This command sends ICMP echo requests (ping), TCP SYN packets to port 443, and TCP ACK packets to port 80 to discover live hosts.

```
(kali@vbox)-[~]
$ nmap -PE -PS443 -PA80 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-06 03:03 PDT
Nmap scan report for 192.168.56.102
Host is up (0.00061s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
445/tcp    open  microsoft-ds
631/tcp    open  ipp
3000/tcp   closed ppp
3306/tcp   open  mysql
8080/tcp   open  http-proxy
8181/tcp   closed intermapper
MAC Address: 08:00:27:E1:D0:8A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 17.83 seconds

(kali@vbox)-[~]
$
```

2. Port Scanning (using Nmap and Netcat):

Port scanning identifies open ports on a target host, indicating active services. Nmap (Comprehensive Scan).

Code: `nmap -sV -p- <target_IP>`

```
(kali@vbox)-[~]
$ nmap -sV -p- 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-06 03:10 PDT
Nmap scan report for 192.168.56.102
Host is up (0.00066s latency).
Not shown: 65524 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          ProFTPD 1.3.5
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7
445/tcp    open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
631/tcp    open  ipp          CUPS 1.7
3000/tcp   closed ppp
3306/tcp   open  mysql        MySQL (unauthorized)
3500/tcp   open  http         WEBrick httpd 1.3.1 (Ruby 2.3.8 (2018-10-18))
6697/tcp   open  irc          UnrealIRCd
8080/tcp   open  http         Jetty 8.1.7.v20120910
8181/tcp   closed intermapper
MAC Address: 08:00:27:E1:D0:8A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: 127.0.0.1, METASPLOITABLE3-UB1404, irc.TestIRC.net; OSs: Unix, Linux; CPE: cpe:/o:linux:lin
ux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 125.12 seconds

(kali@vbox)-[~]
$
```

This command performs a version detection scan on all ports, identifying open ports and the services running on them. Netcat (Basic Port Scan).

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Code: nc -zv <target_IP> <start_port>-<end_port>

```
(kali@vbox)-[~]
$ nc -zv 192.168.56.102 1-9090
192.168.56.102: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.56.102] 9090 (?) : Connection timed out
(UNKNOWN) [192.168.56.102] 9089 (?) : Connection timed out
(UNKNOWN) [192.168.56.102] 9088 (?) : Connection timed out
(UNKNOWN) [192.168.56.102] 9087 (?) : Connection timed out
(UNKNOWN) [192.168.56.102] 9086 (?) : Connection timed out
^C

(kali@vbox)-[~]
$ nc -zv 192.168.56.102 80-8080
192.168.56.102: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.56.102] 8080 (http-alt) open
(UNKNOWN) [192.168.56.102] 8079 (?) : Connection timed out
(UNKNOWN) [192.168.56.102] 8078 (?) : Connection timed out
^C

(kali@vbox)-[~]
$ nc -zv 192.168.56.102 8080
192.168.56.102: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.56.102] 8080 (http-alt) open

(kali@vbox)-[~]
$ nc -zv 192.168.56.102 80
192.168.56.102: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.56.102] 80 (http) open

(kali@vbox)-[~]
```

3. Service Enumeration (using Nmap and Wireshark):

Service enumeration gathers detailed information about the services running on open ports. Nmap (Service Version Detection and Scripting).

Code: nmap -sV -O --script=banner <target_IP>

```
~$ nmap -sV -O --script=banner 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-06 03:50 PDT
Nmap scan report for 192.168.56.102
Host is up (0.00076s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          ProFTPD 1.3.5
|_ banner: 220 ProFTPD 1.3.5 Server (ProFTPD Default Installation) [192.16
|_ 8.56.102]
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ banner: SSH-2.0-OpenSSH_6.6.1p1 Ubuntu-2ubuntu2.13
80/tcp    open  http         Apache httpd 2.4.7
|_ http-server-header: Apache/2.4.7 (Ubuntu)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
631/tcp   open  ipp          CUPS 1.7
|_ http-server-header: CUPS/1.7 IPP/2.1
3000/tcp  closed ppp
3306/tcp  open  mysql        MySQL (unauthorized)
|_ banner: G\x00\x00\x00\xffj\x04Host '192.168.56.101' is not allowed to c
|_ onnect to this MySQL server
8080/tcp  open  http         Jetty 8.1.7.v20120910
|_ http-server-header: Jetty(8.1.7.v20120910)
8181/tcp  closed intermapper
MAC Address: 08:00:27:E1:D0:8A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Aggressive OS guesses: Linux 3.2 - 4.14 (98%), Linux 3.8 - 3.16 (98%), Linux 3.10 - 4.11 (94%), Linux 3.13 - 4.4 (
94%), Linux 3.13 (94%), Linux 3.13 - 3.16 (94%), OpenWrt Chaos Calmer 15.05 (Linux 3.18) or Designated Driver (Lin
ux 4.1 or 4.4) (94%), Linux 4.10 (94%), Android 5.0 - 6.0.1 (Linux 3.4) (94%), Android 8 - 9 (Linux 3.18 - 4.4) (9
4%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Hosts: 127.0.0.1, METASPLOITABLE3-UB1404; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 47.88 seconds
```

4. Scan host systems and find live hosts

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```
kali@vbox:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.395 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.932 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.848 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.758 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=1.13 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=0.716 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=1.07 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=1.04 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=1.04 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=64 time=1.17 ms
^C
--- 192.168.56.102 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9041ms
rtt min/avg/max/mdev = 0.395/0.909/1.165/0.223 ms

(kali@vbox)-[~]
```

```
rtt min/avg/max/mdev = 0.395/0.909/1.165/0.223 ms

(kali@vbox)-[~]$ nmap -sn 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-07 03:16 PDT
Nmap scan report for 192.168.56.101
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 13.01 seconds

(kali@vbox)-[~]$ nmap -sn 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-07 03:16 PDT
Nmap scan report for 192.168.56.102
Host is up (0.00057s latency).
MAC Address: 08:00:27:18:E7:08 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 13.08 seconds

(kali@vbox)-[~]
```

```
192.168.56.101 ping statistics ---
43 packets transmitted, 43 received, 0% packet loss, time 42123ms
rtt min/avg/max/mdev = 0.355/0.693/1.204/0.199 ms
vagrant@metasploitable3-ub1404:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.390 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.845 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.976 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.563 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=0.426 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=0.719 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=0.792 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=0.467 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.383 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=0.509 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=0.679 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=0.399 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=0.687 ms
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=0.793 ms
^C
--- 192.168.56.101 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13040ms
rtt min/avg/max/mdev = 0.383/0.616/0.976/0.187 ms
```

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5. Scan available ports

```
(kali@vbox)-[~]
$ nmap 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-07 03:18 PDT
Nmap scan report for 192.168.56.102
Host is up (0.00059s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
445/tcp    open  microsoft-ds
631/tcp    open  ipp
3000/tcp   closed ppp
3306/tcp   open  mysql
8080/tcp   open  http-proxy
8181/tcp   closed intermapper
MAC Address: 08:00:27:18:E7:08 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 17.88 seconds
(kali@vbox)-[~]
$
```

6. Scan all TCP ports

```
(kali@vbox)-[~]
$ nmap -p- 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-07 03:21 PDT
Nmap scan report for 192.168.56.101
Host is up (0.000020s latency).
All 65535 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 65535 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 13.57 seconds
(kali@vbox)-[~]
$
```

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7. Nmap for network discovery

```
(kali@vbox)-[~]
$ nmap -v 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-07 03:22 PDT
Initiating Parallel DNS resolution of 1 host. at 03:22
Completed Parallel DNS resolution of 1 host. at 03:22, 13.01s elapsed
Initiating SYN Stealth Scan at 03:22
Scanning 192.168.56.101 [1000 ports]
Completed SYN Stealth Scan at 03:22, 0.03s elapsed (1000 total ports)
Nmap scan report for 192.168.56.101
Host is up (0.0000030s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 13.10 seconds
Raw packets sent: 1000 (44.000KB) | Rcvd: 2000 (84.000KB)

(kali@vbox)-[~]
$
```

8. Nmap for scanning and security auditing

```
valid_lft forever preferred_lft forever

(kali@vbox)-[~]
$ nmap -p- --reason 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-07 03:25 PDT
Nmap scan report for 192.168.56.101
Host is up, received localhost-response (0.0000030s latency).
All 65535 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 65535 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 13.60 seconds

(kali@vbox)-[~]
$
```

